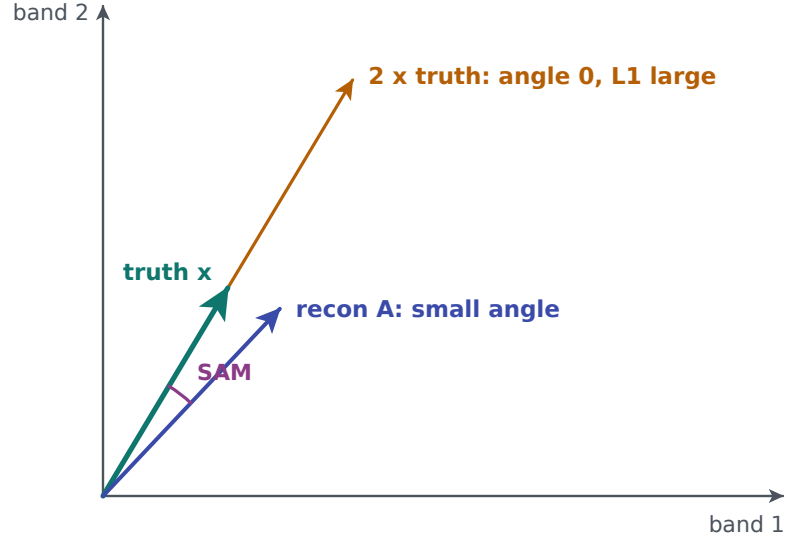
a Two bands of a spectrum, as vectors



Schematic, two bands; the numbers on the right are part 24's three-band example.

b Part 24's numbers, three bands

reconstruction	L1	SAM
A = [0.120, 0.180, 0.330]	0.0233	0.0935 rad
2 x truth = [0.2, 0.4, 0.6]	0.200	0 exactly

truth x = [0.100, 0.200, 0.300]

L1: mean absolute difference per band - it sees brightness.
SAM: the angle between the vectors - scaling a spectrum (shadow, haze, a brighter cloud) does not change it.

SAM = arccos(x . x_hat / (|x| |x_hat|))

The loss uses both, so the model must get the level and the shape right.